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Multiple access protocol:- Multiple access protocol is a set of rules that allow multiple devices to share the same communication channel. When many devices try to send data at the same time their data may collide and the receiver may not receive it correctly. Multiple access protocol help devices share the channel in an organized way and reduce communication problem. Multiple access protocols are divided into three main types:-

1) Random access.
2) Controlled access.
3) Channelization.

Random Access protocols:- Random access protocols allow devices to send data whenever they need to without asking a central controller for permission. All devices have an equal opportunities to use the channel but if two or more devices transmit at same time a collision may occur. When a collision happen the devices usually wait for random period and try to send their data again. The main random access protocols are:-

1) ALOHA 2) CSMA.
3) CSMA/CD 4) CSMA/CA

1) ALOHA:- ALOHA is a simple random access protocol in which devices send data through a shared communication channel. If the receiver get the data successfully it sends an

acknowledgment. If the sender does not receive an acknowledgment within an specific time called timeout, it assume that a collision may have occured. The sender then waits for a random amount of time called backoff time and retransmits the data. If the maximum number of attempts is reached, the sender stops trying.

Pure ALOHA:- In pure ALOHA a device can transmit data at any time without checking whether the channel is busy or free. Because device can transmit at any time the chance of collision is high. If two devices send data during overlapping periods their frames may collide and need to be sent again. The vulnerable time of Pure ALOHA is twice the frame transmission time written as $2T_{fr}$. Its maximum efficiency is approximately 18.4% which means that only small portion of channel capacity is used successfully

Slotted ALOHA:- Slotted ALOHA improves pure ALOHA by dividing time into equal slots. A device can send data only at the beginning of a time slot. This reduces the chances of collision because frames start at fixed times. However, devices must be synchronized with time slots. The vulnerable time of Slotted ALOHA is one frame transmission time written as T_{fr}. Its maximum efficiency is approximately 36.8% making it more efficient than pure ALOHA.

4 Carrier sense multiple access (CSMA):-

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access is a protocol in which a device listen to communication channel before sending data. If channel is free the device transmit. If channel is busy the device waits until it becomes free. CSMA reduces collisions compared with ALOHA but it cannot completely prevent them. This is because two device may find the channel free at nearly the same time and start transmitting before either devices signal reaches the other. The vulnerable time in CSMA is related to propagation time Tp which is the time a signal take to travel through channel.

Types of CSMA persistence methods:-

1) **1-persistent CSMA:-** In 1-persistent CSMA a device continuously checks the channel. As soon as channel becomes free it send data immediately. This method gives a devices a good chance to transmit quickly but collision can occur if multiple devices are waiting.

2) **Non-persistent CSMA:-** In non-persistent CSMA a device checks the channel and if it busy waits for a random amount of time before checking again. This reduce collision but the channel remain free while devices are waiting which can waste time.

3) **P-persistent CSMA:-** In p-persistent CSMA the channel is divided into time slots. When the channel is free a device send data with probability p. With probability 1-p it waits for the next time slot and tries again according to same rule. This method balances the chance of transmission

with the need to reduce collisions.

CSMA/CD:-

CSMA/CD stands for carrier sense multiple access with collision detection. In this protocol a device first listens to the channel before transmitting. While sending data it also checks whether a collision has occured. If a collision is detected the device stops transmitting and sends a jamming signal to inform other devices. It then waits for a random backoff period before trying again. If the maximum number of attempts is reached it stops retransmitting. CSMA/CD was used in traditional shared Ethernet networks. To detect a collision reliably the frame must take enough time to transmit for sender to detect a collision signal returning from the far end of the network. Therefore the frame transmission time must be at least twice the propagation time written as $T_{fr} \geq 2T_p$.
In CSMA/CD an idle channel has no signal energy, normal energy indicates transmission and abnormal energy may indicate a collision. The main purpose of CSMA/CD is to detect collision and respond to them quickly.

CSMA/CA:-

Carrier sense multiple access with collision avoidance. It is mainly used in wireless network and tries to prevent collisions before sending data. A device checks whether the

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the channel is free. If it is busy the device waits if it is free the device wait's for specific period and choose a random backoff time before transmitting CSMA/CA uses an acknowledgment to confirm successful data delivery. Interframe space is the waiting period between transmissions while the contention window provides a range for choosing a random backoff time. If the channel become busy the countdown pauses and resumes when it is free.

It may also use Request to send (RTS) and Clear to send (CTS) before sending data. The Network allocation vector tells other devices how long the channel is expected to remain busy so they can wait. RTS/CTS helps reduce the hidden-station problem which occurs when two devices cannot hear each other but can both send data to the same receiver causing a collision

Controlled access protocols:-

Controlled access protocol allow devices to use a shared communication channel in an organized way. A device cannot simply transmit whenever it wants it must wait for its turn or receive permission. This helps prevent collisions because devices coordinate their transmission The three main controlled access protocols are:-

i) ~~Access protocols~~ Reservation ii) Polling

iii) Token passing.

1) **Reservation protocols:-** In reservation protocol devices first reserve their turns before sending actual data. Time is divided into intervals and each interval begins with reservation period. During this period devices indicates whether they way to transmit. The reservation period may contain same bare slots called mini slots. with a separate mini slots assigned to each device. After making reservation the devices transmit their data according to the reserved order. This method helps organize communication and reduce collisions between data frames.

2) **Polling protocols:-** Polling is controlled access method in which one device called primary station controls communication with other devices called secondary stations. The primary station decides which device can transmit and when. Polling has two main function:-

1) Select 2) Poll.

Select function:- In select function the primary station wants to send data to secondary station. It sends a selection message containing the destination address. The secondary station acknowledges that is ready and then the primary station sends the data. The receiver sends an acknowledgment after receiving the data.

Poll function:- In poll function the primary station ask secondary stations one by one whether they have data to send. If a station has no data it may respond with NAK or another indication that it has nothing to send. If it has

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send data. it sends the data to primary station which acknowledges the transmission.
The main advantage of polling is that it organizes communication and prevents devices from transmitting whenever they want. However polling creates extra communication overhead because the primary station must contact the secondary stations. The primary station can also become a bottleneck and it it fails communication may be disrupted.

B) Token passing protocols:-

Token passing is a controlled access protocol in which a special message called a token circulates among devices. Only the device holding the token can send data. After transmitting it passes the token to the next device.
Its main advantage is that it prevent collisions and provides organized access to the channel.
Its disadvantages are that devices may have to wait for the token and a lost token can disturb communication.
Token passing networks have four arrangements:-

1) In physical ring devices are connected in a circular structure.
2) A dual ring uses two rings.
3) In bus ring or token bus devices are physically connected to bus but follow a logical ring order for token passing.
4) In star ring devices are physically connected around a central device while token passing follows a logical ring.

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Physical addressing (MAC address) :-

Physical addressing is used to identify a network interface on Ethernet network. The physical address is called MAC address. MAC address is 48 bits long and is normally represented by 12 hexadecimal digits. Each hexadecimal digit represents four bits so 12 digits represent 48 bits.

MAC address is divided into two 24-bit portions. The first 24 bits identify the Organizationally Unique Identifier (OUI) which represents an address block assigned to an organization. The remaining 24 bits identify an individual network interface within that block. The IEEE system is responsible for assigning OUIs. The MAC address is used for local Ethernet frame delivery while the IP address is used for logical addressing and communication between networks.

IP addressing:- IP address identifies a devices logically within an IP network. Router's use IP addresses to forward packets between different network, while Ethernet uses MAC address to forward frames within a local network.

IPv4:- IPv4 uses a 32 bit address commonly written in dotted decimal format such as 192.168.1.10. It consists of four 8 bit sections each represented by decimal number from 0 to 255.

IPv6:- IPv6 uses 128-bit address commonly written in hexadecimal group separated by colons such as 2001:0db8::1. IPv6 provides a much larger address space than IPv4.

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Layers of Network:- The three network layers are access layer, distribution layer and core layer. The three layers form a hierarchical network design. Each layer has separate responsibility.

1) Access layer:- The access layer connects end-user devices such as computers, printers to network. Devices such as hubs, switches and wireless access points provide connectivity at this layer. It is layer closest to the end user.

2) Distribution layer:- The distribution layer connects multiple local networks and controls traffic between them. Routers are important devices at this layer because they forward packets between networks. The distribution layer can also provide traffic filtering, security, broadcast containment, logical grouping of user.

3) Core layer:- The core layer provides high speed connectivity between distribution-layer networks. Its main purpose is to transport large amount of network traffic efficiently. Because it carries traffic from multiple parts of an organization it generally require high-capacity and reliable networking equipment.

Hub:- A hub is a networking device that connects multiple devices in an Ethernet network. When a signal arrives at one port the hub repeats it through its other ports. It does not examine ~~[illegible]~~ Consequently all connected devices receives the repeated signal but only the intended normally processes the frame.

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Switch and point-to-point connection:- A switch is a networking device that connects computers and other devices within a local network. It learn the MAC addresses of connected devices and save them in MAC address table. When a switch receives data it checks the destination MAC address and sends the data to the correct port.

Point-to-point connection:- Point-to-point is the direct connection between two devices. **E.g:-** when a computer is connected to switch through an Ethernet cable. It form a point to point connection.

If the switch does not know the MAC address it send the frame through all its other ports except the port where it received the frame. This is called flooding. Once the switch learn the destination MAC address it can send future frames to the correct port.

Broadcasting:- Broadcasting means sending a message from one device to all devices on some local network. In Ethernet the broadcast MAC address is FF-FF-FF-FF-FF-FF. It is 48 bits long. Each hexadecimal F represents four binary ones. **E.g:-** when a computer wants to find the MAC address for another computer it sends an ARP request to all devices on local network. Switches forward broadcast messages to other ports in same broadcast domain while routers normally do not forward local broadcast to other networks.

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Unicast:- Unicasting means sending a message from one device to one specific device.

Example:- When computer A send file to computer B it is unicast communication. A switch uses the destination MAC address to send the data to correct port

Multicasting:- Multicasting means sending a message from one device to selected group of devices not to everyone.

Example:- a teacher sends a live video stream to a specific group of students computer. This is multicast communication if the computers receive the stream through a multicast group

Switch and MAC address:- A switch is better than a hub because it can learn MAC addresses and send data to correct port. It create separate connections for connected devices which improves network performance. A switch maintains a MAC address table that stores the MAC addresses of devices and the switch ports connected to them. This table helps the switch decide where to send an Ethernet frame. The switch stores its MAC address table in memory, commonly RAM. The entries have a limited life time, meaning they are removed if they are not refreshed for a certain period. The switch updates the table dynamically by learning the source MAC addresses of incoming frames. If the switch does not know the destination MAC address, it uses a process called flooding. It sends the frame through all the other ports except the port where the frame arrived. When the destination address is known the switch can send the frame to correct port.

ARP (Address resolution protocol.) :- ARP stand for address resolution protocol. It is used to find the MAC address associated with known IPv4 address on local network. Example: Computer A knows the IP address of Computer B but does not know its MAC address. Computer A uses ARP to discover Computer B's MAC address. After receiving the answer Computer A stores the IP-to-MAC mapping in its ARP table so it can communicate with Computer B.

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Three steps of ARP:-

ARP request:- The sender broadcasts an ARP request asking "Who has this IP address?" All devices on local network receive the request.

IP address Match:- Each device checks the requested IP address. The device whose IP address matches the request prepares an ARP reply containing its MAC address.

ARP reply and storage:- The matching device sends its MAC address to sender. The sender stores the IP-to-MAC mapping in its ARP table and can then send data to destination.